

```

In [3]: library(Seurat)
# library(SeuratDisk)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
library(clustree)
library(repr)
options(repr.plot.width=10, repr.plot.height=8)

library(UpSetR)
library(grid)

library(PRRoc)
library(Matrix)

getwd()

dataset_id <- "simulated_mm_RA"
genome_id <- "mm10"
samples <- c("all")

for (sample in samples) {
  dir.create(paste0("figures_", dataset_id, "_", sample))
}

colorTools <- c( # "MATES"="#F98A7B",
  "STARsolo_TE" = "#FFBC81",
  "STARsolo_TE_EM" = "#F3E088",
  "SoloTE_unique" = "#A4DD9B",
  "SoloTE_thr2" = "#6FC69D",
  "SoloTE_thr1" = "#4BB2BB",
  "SoloTE_thr0" = "#3989BF",
  "Stellarscope" = "#4F5D93",
  "simulated" = "grey70"
)

thrMinCells <- 500 * 0.05

```

Loading required package: SeuratObject

Loading required package: sp

```
'SeuratObject' was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall 'SeuratObject' as the ABI for R may have changed
```

```
'SeuratObject' was built with package 'Matrix' 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall 'SeuratObject' as the ABI for 'Matrix' may have changed
```

```
Attaching package: 'SeuratObject'
```

```
The following objects are masked from 'package:base':
```

```
intersect, t
```

```
Attaching package: 'anndata'
```

```
The following object is masked from 'package:SeuratObject':
```

```
Layers
```

```
Warning message:
```

```
"package 'dplyr' was built under R version 4.4.3"
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
Warning message:
```

```
"package 'tidyr' was built under R version 4.4.3"
```

```
Warning message:
```

```
"package 'RColorBrewer' was built under R version 4.4.3"
```

```
Loading required package: ggraph
```

```
Attaching package: 'ggraph'
```

```
The following object is masked from 'package:sp':
```

```
geometry
```

```
Loading required package: rlang
```

```
Warning message:
```

```
"package 'rlang' was built under R version 4.4.3"
```

```
Attaching package: 'Matrix'
```

```
The following objects are masked from 'package:tidyr':
```

```
expand, pack, unpack
```

```
'/mnt/TEdata/TEbenchmarking/Rscripts'
```

```
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):  
"figures_simulated_mm_RA_all" already exists"
```

```
In [1]: # import workspace  
load("workspaces/00_evaluation_objectCreation.Rdata")  
sample <- "all"
```

Evaluation

Metrics by age

```
In [4]: layer <- "counts"

matSplatter <- GetAssayData(splatter_objs[[sample]], layer=layer)

# split the matrix by age
splatter_age_vec <- conversion_table$ageClass[match(rownames(matSplatter), conversion_table$stellarscopeID)]

matSplatter_age_List <- list()
matSplatter_age_List[["old"]] <- matSplatter[splatter_age_vec=="old",]
matSplatter_age_List[["young"]] <- matSplatter[splatter_age_vec=="young",]

precisionTool_age_List <- list()
recallTool_age_List <- list()

for(age in c("old", "young")){

  precisionTool_age_List[[age]] <- list()
  recallTool_age_List[[age]] <- list()

  for(tool in names(objList)){

    # get matrix of the tool, keep only TEs of the right age
    matTool <- as.matrix(GetAssayData(objList[[tool]][[sample]], layer=layer))

    #print(table(rownames(matTool) %in% conversion_table$stellarscopeID))

    tool_age_vec <- conversion_table$ageClass[match(rownames(matTool), conversion_table$stellarscopeID)]
    matTool <- matTool[tool_age_vec==age,]

    precisionTool_age_List[[age]][[tool]] <- NULL
    recallTool_age_List[[age]][[tool]] <- NULL

    for(cell in Cells(splatter_objs[[sample]])){

      exprSplatter <- names(matSplatter_age_List[[age]][,cell])[matSplatter_age_List[[age]][,cell] > 0] # TEs e

      exprTool <- names(matTool[,cell])[matTool[,cell] > 0] # TEs detected by the tool

      TP <- intersect(exprSplatter, exprTool)
      nTP <- length(TP)

      FP <- setdiff(exprTool, exprSplatter)
      nFP <- length(FP)

      FN <- setdiff(exprSplatter, exprTool)
      nFN <- length(FN)

      precision <- nTP / (nTP + nFP)
      precisionTool_age_List[[age]][[tool]] <- c(precisionTool_age_List[[age]][[tool]], precision)

      recall <- nTP / (nTP + nFN)
      recallTool_age_List[[age]][[tool]] <- c(recallTool_age_List[[age]][[tool]], recall)

    }
  }
}
```

Plot F1 score

```
In [ ]: options(repr.plot.width=5, repr.plot.height=3)

meanF1score_age_df <- NULL

for(age in c("old", "young")){

  precisionDf <- stack(precisionTool_age_List[[age]])
  colnames(precisionDf) <- c("precision", "tool")

  recallDf <- stack(recallTool_age_List[[age]])
  colnames(recallDf) <- c("recall", "tool")

  df <- merge(precisionDf, recallDf, by='tool')

  df$F1score <- (2 * df$precision * df$recall) / (df$precision + df$recall)

  df$tool <- factor(df$tool, levels=rev(names(colorTools)))
}
```

```

# plot just the mean value

precisionDf <- stack(lapply(precisionTool_age_List[[age]], mean))
colnames(precisionDf) <- c("precision", "tool")

recallDf <- stack(lapply(recallTool_age_List[[age]], mean))
colnames(recallDf) <- c("recall", "tool")

Fscores <- sapply(names(precisionTool_age_List[[age]]), function(tool){
  (2 * precisionTool_age_List[[age]][[tool]] * recallTool_age_List[[age]][[tool]]) /
  (precisionTool_age_List[[age]][[tool]] + recallTool_age_List[[age]][[tool]])
})
meanFscores <- colMeans(Fscores)
FscoreDf <- stack(meanFscores)
colnames(FscoreDf) <- c("F1score", "tool")

df <- merge(precisionDf, recallDf, by=c("tool"))
df <- merge(df, FscoreDf, by=c("tool"))

df$tool <- factor(df$tool, levels=rev(names(colorTools)))

# create df for facet plot
meanF1score_age_df <- rbind(meanF1score_age_df, cbind(df, age))
}

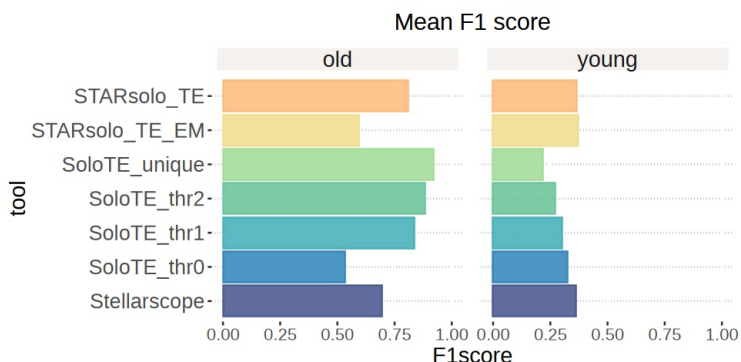
```

In [7]: `options(repr.plot.width=8, repr.plot.height=4)`

```

ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 2) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean F1 score")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=16),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))
ggsave(paste0("figures_", dataset_id, "_", sample, "/meanF1score_tool_byAge_horiz.png"), device="png", width=8, height=4)
ggsave(paste0("figures_", dataset_id, "_", sample, "/meanF1score_tool_byAge_horiz.pdf"), , height=3.2, width=8)

```



In [8]: `options(repr.plot.width=5, repr.plot.height=6)`

```

ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean F1 score")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=18),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))
ggsave(paste0("figures_", dataset_id, "_", sample, "/meanF1_tool_byAge.png"), device="png", width=5, height=6)
ggsave(paste0("figures_", dataset_id, "_", sample, "/meanF1_tool_byAge.pdf"), device="pdf", width=5, height=6)

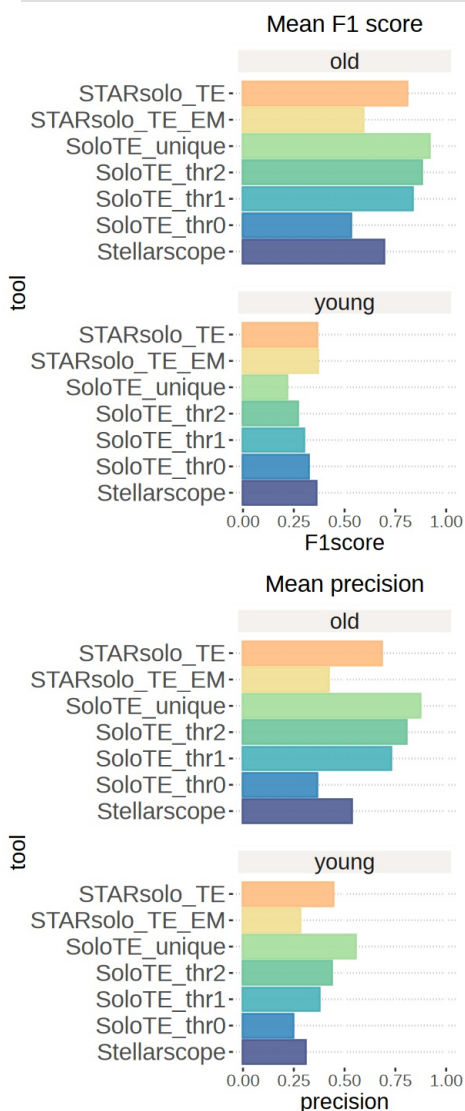
```

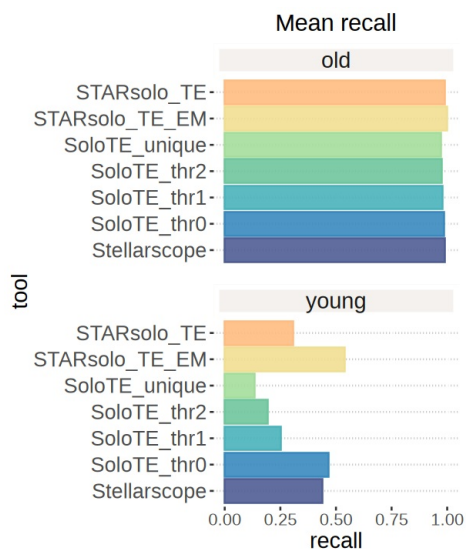
```

ggplot(meanF1score_age_df, aes(x=precision, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean precision")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=18),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))
ggsave(paste0("figures_",dataset_id,"_",sample,"/meanPrecision_tool_byAge.png"), device="png", width=5, height=6)
ggsave(paste0("figures_",dataset_id,"_",sample,"/meanPrecision_tool_byAge.pdf"), device="pdf", width=5, height=6)

ggplot(meanF1score_age_df, aes(x=recall, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean recall")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=16),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))
ggsave(paste0("figures_",dataset_id,"_",sample,"/meanRecall_tool_byAge.png"), device="png",width=5, height=6)
ggsave(paste0("figures_",dataset_id,"_",sample,"/meanRecall_tool_byAge.pdf"), device="pdf",width=5, height=6)

```



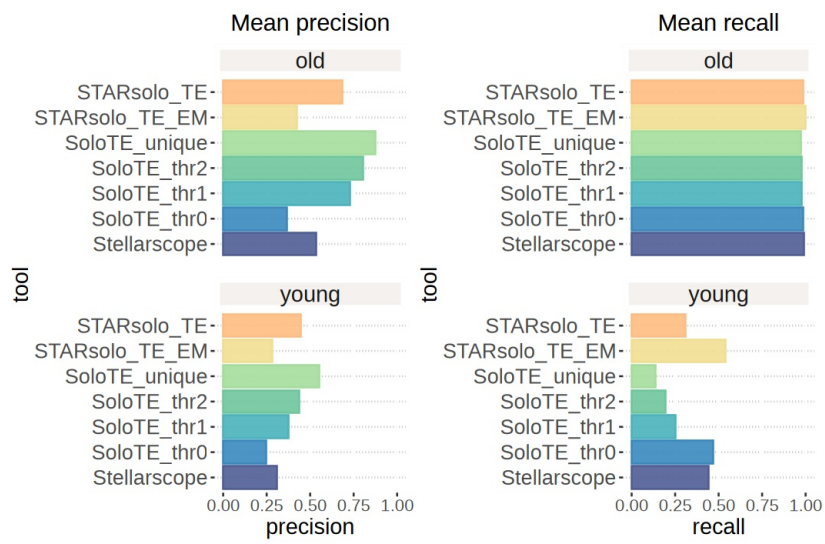


```
In [9]: options(repr.plot.width=9, repr.plot.height=6)

precision_plot <- ggplot(meanF1score_age_df, aes(x=precision, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean precision")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=16),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))

recall_plot <- ggplot(meanF1score_age_df, aes(x=recall, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean recall")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=16),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))

precision_plot + recall_plot
ggsave(paste0("figures_", dataset_id, "_", sample, "/meanPrecisionRecall_tool_byAge.png"), device="png", width=9, height=6)
ggsave(paste0("figures_", dataset_id, "_", sample, "/meanPrecisionRecall_tool_byAge.pdf"), device="pdf", width=9, height=6)
```



```
In [10]: write.table(meanF1score_age_df, paste0("data/", dataset_id, "_meanF1score_age_df_mixed.tsv"))
```